

Team/Project Name: ToxOracle

Narrative (how we want to “sell” it to the judges/people)

Overarching goal: Predicting whether a drug will be toxic in humans from preclinical data and help resolve discrepancy between in vitro assays and the ‘real’ toxicity in human trials.

What the data input could be: We can get data on animal models / 2D and even 3D models along with drug structures (i.e. SMILES) and in vivo data (from human trials) as an input.

What the demo could look like: A researcher could load on new SMILES structures to get an understanding of toxicity in vitro and potential toxicity directly in humans → along with discrepancy score (if the tox i will get in vitro sufficient for human tox or not?)

The Story Line:

Drug development is costly, time consuming and inefficient: too many drugs are still failing. One of the leading causes of drug attrition is toxicity. Especially toxicity that is not captured in *in vitro* assays which appears in human trials. Liver toxicity is the number 1 drug toxicity event in the clinic. Can we build an agent that would allow scientists, from a drug structure, to understand whether the drug will actually be toxic in humans (predicting from the pre-clinical data the actual / true human toxicity).

This could be used by scientists early in preclinical tox screening to understand which drug candidate to prioritise, and which studies (i.e. suggesting 3D organoids) are required.

We have used NVIDIA BioNeMo and Rosalind in Codex to build this, using FDA labelled drug-induced liver injury drug annotation for true human toxicity (FDA DILI 2.0) and in vitro dataset exposure and viability.

We have made a novel model using novel FDA DILI 2.0 to predict drug toxicity for the liver. The next step would be expand this to over organs to better understand

Dataset

Model Architecture

Overall outline

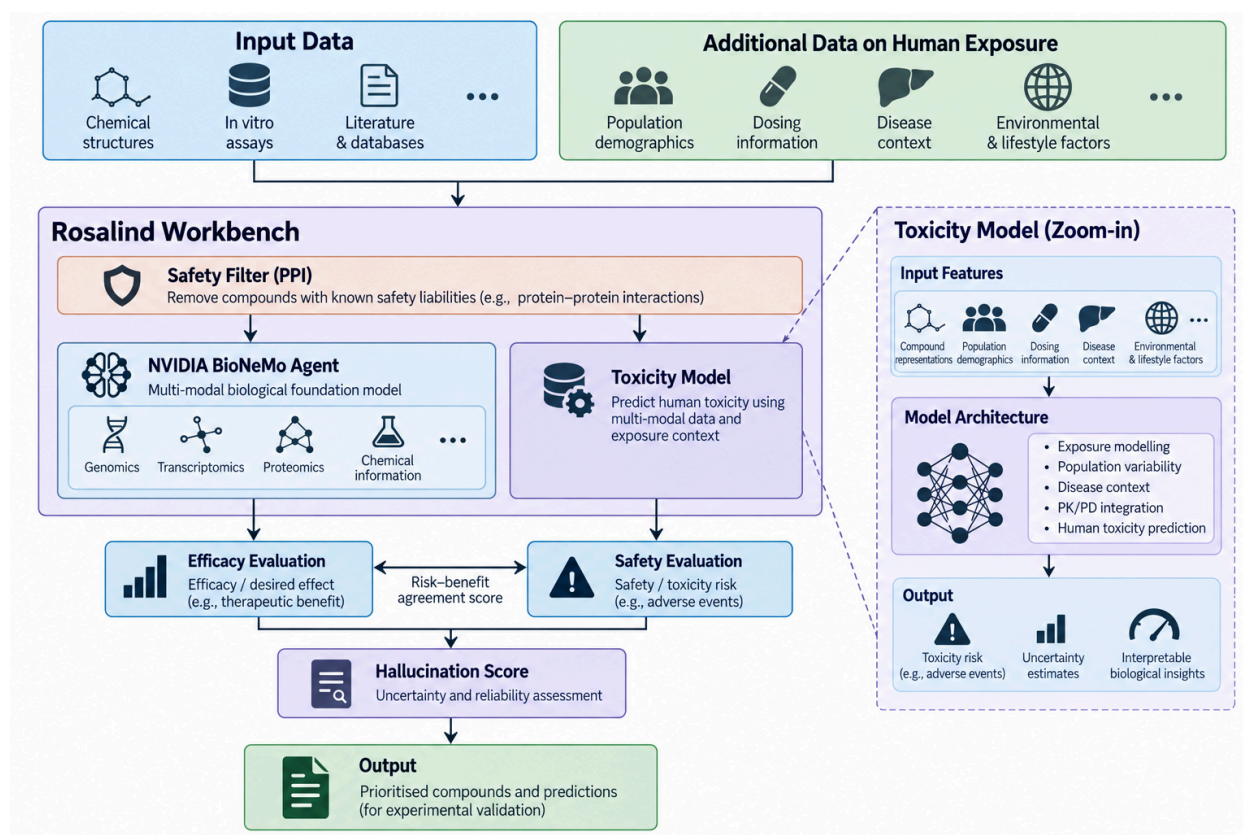
We will build an agentic drug-discovery model that supports early-stage candidate identification, paired with a dedicated toxicity model designed to flag compounds that appear promising initially but are at risk of failing later in human studies. By identifying these potential downstream failures earlier, the system could reduce wasted experimental effort and lower the overall cost of the drug-development pipeline.

To make richer human data usable without compromising privacy, we also propose a separate safety and privacy layer that prevents subject-sensitive information from being transmitted to external big-tech model providers. Importantly, this layer should be isolated from OpenAI, NVIDIA, or other third-party model functionality. Researchers could then incorporate sensitive human-exposure data locally and securely, enabling more biologically and clinically informed drug-discovery decisions without exposing identifiable or restricted subject-level information.

An Idea to Keep In Mind

Konstantine talked about the “intelligence per token” idea. Maybe we can incorporate token efficiency into our design.

General schematics



Decision Points

- Agent design
 - From Rosalind workbench to NVIDIA BioNeMo agents and underlying models
 - UX: how the users (biologists) will experience our whole product
- Models
 - Which BioNeMo and toxicity (e.g., ToxPredictor) model we will use

- Agent workflow
- Evaluation metrics
 - All the conventional metrics for drug candidate identification
 - Conventional toxicity metrics
 - Disagreement score highlighting how the drug candidate prediction will perform at the initial screening stage and future human-level clinical studies
 - (+) If we have time after building MVP, hallucination scores, safeguarding whether the candidates we suggest are not a product of some hallucinated noise
- Output
 - In relations to demo webapps and visualisations